

SECTION B – WRITTEN QUESTIONS (Total: 80 marks)

Answer **ALL** questions in this section. Marks are indicated at the end of each question. Together they are worth 80% of the total marks for this examination.

Question 1 (20 marks – approximately 36 minutes)

Alder Investment ("Alder") is considering investing in the stock market. The Chief Financial Officer ("CFO") of Alder has picked three stocks from different industries listed on the Hong Kong Stock Exchange with the following characteristics:

	<u>Expected Return</u>	<u>Standard Deviation</u>
Stock A	10%	2.0%
Stock B	13%	2.2%
Stock C	16%	3.0%

The expected return on the market is 12%, and the rate of Hong Kong Exchange Fund Bills and Notes is 3%.

Required:

- (a) Compute the beta coefficient for each of the three stocks identified by the CFO.
(3 marks)
- (b) Calculate the coefficient of variation for each stock.
(3 marks)
- (c) Alder is considered as a risk-averse investor. Based on the answers in Question 1(b), identify and explain which stock should be selected if Alder plans to invest in one stock only.
(2 marks)
- (d) As the CFO believes that it is never a good idea to rely on one investment, Alder has decided to invest in a portfolio with equal weight in all the three stocks.
 - (i) Compute the expected return of this portfolio.
(2 marks)
 - (ii) Explain whether the total risk of this portfolio can be measured by the weighted average of individual stock's standard deviation.
(4 marks)
- (e) Explain the differences between systematic risk and unsystematic risk and identify one example each for these two risks.
(6 marks)

Question 2 (20 marks – approximately 36 minutes)

Butternut Limited ("Butternut") is a trading company in Hong Kong. In the beginning of 2021, the finance department of Butternut prepared the pro forma financial statements for the year ended 31 December 2021. The extracted information is as follows:

Pro forma statement of profit or loss for the year ended 31 December 2021

	HK\$'000
Net sales	2,050
Operating costs	1,630
Depreciation	85
Income before interest and tax	335
Interest	45
Income before tax	290
Tax	49
Net income	241
Dividends	96
Retained profit	145

Pro forma statement of financial position

	<u>2020</u> (Actual) HK\$'000	<u>2021</u> (Forecast) HK\$'000
Non-current assets	1,155	1,180
Current assets		
Inventories	120	152
Accounts receivable	180	165
Cash	220	431
	<u>520</u>	<u>748</u>
Total assets	<u>1,675</u>	<u>1,928</u>
Non-current liabilities	500	500
Current liabilities		
Accounts payable	150	188
Tax payable	42	80
Dividends payable	173	205
Total current liabilities	<u>365</u>	<u>473</u>
Share capital	500	500
Retained earnings	310	455
Total equity	<u>810</u>	<u>955</u>
Total liabilities and equity	<u>1,675</u>	<u>1,928</u>

Other information: Non-current assets do not include any intangible assets and Butternut has no plan to sell any non-current assets in 2021. All interest incurred is paid in the same year.

Required:

- (a) **Prepare a pro forma statement of cash flows for the year ended 31 December 2021 using indirect method.**

Note: The format of the pro forma statement of cash flows is not required to follow Hong Kong Accounting Standard 7.

(12 marks)

- (b) **Compute the forecast return on equity for Butternut for the year ended 31 December 2021 by multiplying the three component parts of DuPont equation.**

(4 marks)

- (c) **Explain how the DuPont equation can be used by the management of a company.**

(4 marks)

Question 3 (20 marks – approximately 36 minutes)

Cottonwood Investment Limited ("Cottonwood") is considering investing in a new project which will generate an annual cash flow of HK\$1.2 million in perpetuity. The initial investment amount for this project is HK\$10 million. Cottonwood decided to raise the HK\$10 million capital required as a combination of debt, preference shares and ordinary shares. The breakdown is as follows:

Debt: HK\$5 million debt will be raised by borrowing from a bank at a fixed rate of 6% per annum.

Equity (preference shares): HK\$3 million preference shares which will pay annual dividend of 8%.

Equity (ordinary shares): HK\$2 million equity will be raised from an issue of ordinary shares.

The following information related to Cottonwood is provided:

- Estimate of growth in dividends = 6% per year in perpetuity
- Last dividend just paid = HK\$2.5 per share
- Current market price of outstanding ordinary shares = HK\$30 per share
- Tax rate = 16.5%

Required:

- (a) Compute the cost of equity (ordinary shares) for Cottonwood using the dividend growth model. (2 marks)
- (b) Using the result in Question 3(a) as the cost of equity (ordinary shares) for the new project, compute the weighted average cost of capital for this project. (3 marks)
- (c) Compute the net present value of the new project and advise whether it should be accepted or not. (3 marks)
- (d) Outline the advantages and disadvantages for Cottonwood in using debt compared to issuing equity to raise additional capital for a new project. (6 marks)
- (e) Explain the pecking order theory and how it is used to determine the capital structure of a company. (6 marks)

Question 4 (20 marks – approximately 36 minutes)

- (a) Daphne Manufacturing Limited ("DML") is an electronic toys manufacturing company in Hong Kong. DML produces three components, X, Y and Z, which are incorporated into different products. All the components are produced using the same specialised machinery. DML also has an option to purchase these components from an external supplier. The in-house production data and the purchase cost from an external supplier are as follows:

	X	Y	Z
Production cost per unit (HK\$):			
Direct material	70	60	30
Direct labour	50	40	28
Variable overhead	24	15	11
Fixed overhead allocation	25	15	10
Machine hours per unit	5	4	1
Number of units required per annum	3,000	1,000	2,000
Purchase cost per unit from the external supplier (HK\$)	200	140	78

Required:

- (i) The maximum number of specialised machine hours available per annum is 15,800. Determine which component/ components and its/ their respective number of units should be purchased from the external supplier. (8 marks)
- (ii) Apart from the cost consideration in Question 4(a)(i), list TWO other factors that DML should consider when choosing an external supplier. (2 marks)

- (b) DML has just made a sale of its products to a customer in Europe. DML agreed to receive the payment in EUR as requested by the customer. According to the sales contract, DML will receive EUR200,000 in three months. DML plans to mitigate the currency risk in this sales transaction by entering into a forward contract to convert EUR200,000 to HK\$. DML obtained the following market information from its bank:

Current spot exchange rate is HK\$8.80 per EUR.

Three-month forward rate is HK\$8.60 per EUR.

Required:

- (i) Assuming that DML has entered into the three-month forward contract but it has not made the overseas sales to Europe, determine the gain or loss in HK\$ on the forward contract if EUR appreciates by 5% against HK\$.
(3 marks)
- (ii) Based on the result in Question 4(b)(i) above, explain why DML should hedge the currency exposure with the forward contract.
(4 marks)
- (iii) Apart from the currency risk, identify another major risk on the EUR200,000 receivable that DML is facing in the next three months. Advise TWO ways that DML can mitigate this risk.
(3 marks)

* * * END OF EXAMINATION PAPER * * *